

Eucalyptus was developed at the Computer Science Department of the University of California at Santa Barbara as a research project, and is now enhanced and maintained by Eucalyptus Systems, a company founded by the original developers.

A UEC cloud consists of one or more nodes, where each node has one node controller. One or more node controllers are managed by a cluster controller, and the entire cloud is managed by a cloud controller. Refer to the box titled 'Eucalyptus and the various controllers' to get an idea about the architecture.

Installing Eucalyptus

This article describes the installation of UEC on a two-machine configuration, in which Node 1 contains the CLC, CC and SC, while Node 2 contains the NC. Other variations include having the CLC, CC, and SC on separate nodes, as well as having multiple NC and multiple nodes.

First, we install the CLC, CC and SC packages on Node 1:

```
$ sudo apt-get install eucalyptus-cloud eucalyptus-cc \
eucalyptus-walrus eucalyptus-sc
```

Next, we install the NC package on Node 2:

```
sudo apt-get install eucalyptus-nc
```

We configure `/etc/eucalyptus/eucalyptus.conf` with the name of the bridge (`virbr0`), and restart the NC.

We install the CLC Eucalyptus user's public SSH key into the NC Eucalyptus user's `authorized_keys` file.

We need to register the CC and SC with the CLC. Since, in this installation, the CC and SC are on the same node as the CLC, the registration is automatic. In case they are on different nodes, the registration is accomplished using the following code:

```
$ sudo start eucalyptus-cc-registration
$ sudo start eucalyptus-sc-registration
$ sudo start eucalyptus-walrus-registration
```

The next step is to register the NCs with the CLC. This is done by adding the IP address of each NC in `/etc/eucalyptus/eucalyptus.conf` of the CLC.

We next need to obtain credentials on the CLC. This is done using the code below:

```
$ sudo euca_conf --get-credentials mycreds.zip
```

The credentials (`unzip mycreds.zip`) are stored in the `.euca` directory in the home of the Eucalyptus user.

Euca2ools are installed using the code below:

```
$ sudo apt-get install euca2ools
```

To check if things are working fine, issue the following command:

```
$ euca-describe-availability-zones verbose
```

This should list the details of the cloud.

Finally, we need to install an Ubuntu Karmic image to run in the virtual machines. This is done by connecting to the following:

<https://<cloud-controller-ip-address>:8443/>

Login for the first time using `admin/admin`, and then change the password. After image creation is complete, the image is ready to run in the VM. Please refer to the UEC package install link in the References section below for more details on the steps to install UEC.

That's all folks—you now have your private cloud ready for use. 

References

- KVM: http://www.linux-kvm.org/page/Main_Page
- Ubuntu KVM install: <https://help.ubuntu.com/community/KVM>
- Ubuntu KVM install Howto: <http://www.howtforge.com/virtualization-with-kvm-on-ubuntu-9.04>
- QEMU: <http://www.qemu.org/>
- libvirt: <http://libvirt.org/>
- Eucalyptus: <http://open.eucalyptus.com/>
- UEC: <https://help.ubuntu.com/community/UEC>
- UEC package install:
- Processors supporting virtualization: http://en.wikipedia.org/wiki/X86_virtualization
- UEC package install: <https://help.ubuntu.com/community/UEC/PackageInstall>

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